

Detecting Deception

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49 DETECTING DECEPTION

Key Ideas

Deception is the deliberate attempt to mislead another person with a communication that is in some way false. Communication may occur across multiple channels including verbal (such as an account of an event), nonverbal (for example, nodding), and paraverbal (including, tone of voice), and any of these channels may include deceptive information. Further, although deception is often thought of as falsification (presenting false information as truthful), it may also involve withholding true information known to be relevant, diversion, exaggeration, and half-truths. The purpose of a deceptive communication is to affect the decision-making of the receiver of the communication, by controlling information to manipulate their cognitive, psychological, or emotional state.

Deception is a ubiquitous behaviour, and most people engage in some form of deceptive communication every day (DePaulo et al., 1996). The majority of this deception is low-stake, 'white lies', the consequences of which are not serious. However, in the world of crime and policing, not only is there is an increased contextual expectation of deception, but in this type of high stakes scenario the consequences of deception (whether successful or unsuccessful) are likely to be significant. Detecting deception, therefore, may be important in several areas, including: interrogation, which is an approach used in several countries (including the USA) that rests on the idea that it is possible to accurately judge the veracity of a suspect, and then employ a process that results in a confession; intelligence gathering, in which the credibility of information may have to be assessed in order to make decisions; investigative interviewing, the function of which in the UK is information-gathering not judging veracity, although it is likely that in practice veracity judgements are made; and jury decision-making, which may in part be based on the perceived veracity of defendants, victims, and witnesses.

The detection of deception (whether by humans or an automated system), is dependent on the premise that there are behaviours, also known as cues, that differ between truthful and deceptive communications. Furthermore, the cues must be

observable in some way (individual's need to be able to perceive them, with or without specialist equipment); the cues should be consistent within and between individuals (the same cues occur for all types of deception for everybody); the cues must be specific to deception (they do not occur for other reasons); and the cues must be sensitive to deception (they always occur in a deceptive communication). The cues should also be robust to countermeasures (they cannot be suppressed or faked), and the cues must be understood by the observer (individual's know that the cues indicate deception). The only known cue that satisfies all these criteria is Pinocchio's nose, which has the obvious limitation of being fictional. There is general agreement amongst researchers that, in reality, an equivalent cue does not exist (van Koppen, 2012). However, there is little else in the deception detection literature upon which there is general agreement. It has been established that there is no strong consensus amongst deception researchers about which cues or approaches to detecting deception are most valuable, and no agreed-upon dominant and unifying theory. This leaves a body of literature on detecting deception that is rather fragmented and inconsistent.

Key Findings and Approaches

Two widely-cited meta-analytic findings suggest that cues to deception tend to be unreliable and weak, and that people are generally able to detect deception barely above the level of chance (including people who regularly encounter deception in their professional lives) (DePaulo & Bond, 2012). These findings seem obviously related, as accuracy in detecting deception necessarily depends on available cues. A meta-analysis exploring the relationship between these findings confirmed that people's intuitive beliefs about cues to deception are rarely incorrect, and that what limits accuracy is the weaknesses of the cues (Hartwig & Bond, 2011). The logical implication of this is that accuracy will be increased by maximising the behavioural differences between truthful and deceptive communications. Related to this is the common call amongst researchers to address issues of ecological validity in the literature: the majority of findings are the result of lab-based, experimental studies in which there are no serious consequences for not being believed. This methodology may be analogous to the experience of producing every day 'white lies', but may not reflect the cognitive, emotional, and motivational correlates of lying in a high stakes-context, such as a police interview (Frank & Svetieva, 2012). Furthermore, calls for

the consideration of context are increasing amongst researchers, as not all high-stakes lies will necessarily resemble each other (for example, a person lying about killing a close relative may be expected to experience some differences in emotion compared to a person lying about insurance fraud), and so cues may vary according to context (van Koppen, 2012).

There are four major approaches to detecting deception; observing non-verbal behaviour, measuring physiological responses, measuring neural activity (sometimes called 'brain mapping'), and analysing accounts (i.e. verbal behaviour). In discussing these, the focus will be directed at approaches which do not require specialist equipment, as they are likely to have greater utility in an applied policing context. It should be noted that, due largely to the substantial threat posed by modern terrorism, the field has developed to include attempts to identify hostile intent. The focus is no longer solely on lying about past actions, but now includes lying about future intentions, with the aim of preventing harmful acts before they occur (Mac Giolla et al, 2015, p.155).

Nonverbal Behaviour

An idea often propagated in the media is that people can spot when people are being deceptive by leakage of their true intent via non-verbal behaviours such as facial expressions, face-touching, fidgeting and gaze aversion. The common assumption is that behaviours which indicate nervousness are related to deception, but there is very little empirical support that any non-verbal behaviours appear consistently in deceptive communications, and the premise that people can identify deception by watching a person's behaviour is not robust (Bond et al, 2015, p. 42-45). Emerging research areas around non-verbal behaviours that may prove more fruitful in the future as they are not based on nervousness but on cognition include ocular responses (Gamer & Pertzov, 2018, p.169-186) and reaction times (Suchotzki, 2018, p.243-268), although as both require specialist equipment, they are not considered here in detail. In the interests of dispelling myths, it should be noted that there is no evidence for the popular proposition related to Neurolinguistic Programming that people look to their upper right when lying (Gamer & Pertzov, 2018, p.171-172).

Physiological Responses

Measuring physiological responses is typically associated with the polygraph (sometimes incorrectly referred to as a lie detector), which measures cardiovascular, respiratory and electrodermal responses, all of which are indicators of arousal of the autonomic nervous system (part of the nervous system that regulates involuntary processes). Arousal is an involuntary physiological response to, amongst other things, anxiety, pain, stress, and situations that are unusual or threatening. The premise is that when people are lying they may experience increased arousal due to the psychological discomfort associated with lying, and this can be measured. The obvious problem with this approach is that arousal is a very generalised response that is not specific to deception, and can have innumerable causes (for example, many people may find a police interview stressful whether they are truthful or deceptive). Nevertheless, polygraphs are used in investigations in several countries, and occasionally in the monitoring of sex offenders post-conviction. The accuracy of the polygraph has not been robustly established despite decades of research, and the results of polygraph exams are generally not admissible as evidence (Ambach & Gamer, 2018, p.24-25).

Due to the importance of modern transit security, and the large-scale screening required, several automated systems have been developed in attempts to detect hostile intent in real time. These systems remotely measure physiological and non-verbal responses including facial temperature, heart and respiration rates, eye movements, facial expressions and body movements, and vocal pitch. Notwithstanding the ethical and privacy concerns around covertly collecting biological data in an attempt to predict people's future intentions, scientists have voiced concerns over the theoretical foundations of these systems which are based largely on the assumption that those with hostile intent will experience greater arousal than others, for which there is no robust evidence (Weinberger, 2010). In sum, although physiological responses can be measured very sensitively, the major limitation of this approach is specificity (i.e. the responses occur for a variety of reasons, not specifically due to deception).

Neural Activity

Measuring neural activity to detect deception will be very briefly discussed as research is in its infancy and specialist equipment is required, so the relevance to

policing is very limited. Overall, there is some evidence that areas of the brain related to performing complex cognitive tasks tend to be activated during deceptive communication, but those areas are also activated during other cognitive tasks, so the issue again is with specificity (Ganis, 2018, p.152-152). A promising strand of research is around the P300 response, based on the principle that a measurable response to meaningful information occurs in the brain 300ms after exposure to the information. Consequently, it can be expected that guilty suspects will produce this response when provided with crime-relevant details, whereas innocent suspects will not. This has the benefit of being outside conscious control and resistant to counter measures, and some studies have reported very high accuracy, although it is yet to be established whether laboratory findings can be replicated in the field (see Rosenfield et al., 2018). Despite this, the P300 response has recently been used as evidence in the prosecution of a murder case in Dubai (Al Shouk, 2021).

Analysing Verbal Accounts

The two most well-established approaches to analysing accounts of events to assess credibility are Statement Validity Analysis (SVA) (Kohnken & Steller, 1988) a core component of which is Criteria-Based Content Analysis (CBCA), and Reality Monitoring (RM) (Johnson & Raye, 1981). These approaches share conceptually similar underpinning theories, which are that genuinely experienced events are recalled and described differently to fabricated events, as people fabricating events have less knowledge and experience of what happened. Both approaches have a broad evidence base, and SVA is admissible as evidence in child sexual abuse cases in several countries (SVA was developed for use in this type of case). There is substantial overlap in the cues identified in these approaches, and meta-analytic findings indicate that the majority of those with the strongest evidential support are cues related to detail and specificity, which are predictive of truthful accounts; contextual embedding, reproduction of conversation, quantity of details, and perceptual, temporal and spatial information (Vrij, 2015, p.12, p.18). Again, in the interests of dispelling myths, Scientific Content Analysis (SCAN), a method of assessing accounts developed by Sapir (http://www.isiscan.com/intro_to_scan.htm) and that is widely used by practitioners (particularly in the United States and Canada), has not been extensively tested and has scant supporting evidence (Oberlander et al, 2021).

Eliciting Cues to Deception

More recently, several tools have been developed which take a more active approach by eliciting cues to deception, rather than passively observing. A commonly-accepted theoretical proposition is that liars experience greater cognitive load than truth-tellers, as they have complex and simultaneous cognitive tasks to perform. Simply put, liars need to think harder than truth-tellers. Consequently, increasing the cognitive load of interviewees should maximise the differences between liars and truth-tellers. Asking unanticipated questions relating to reasonably expected knowledge, and encouraging interviewees to provide more detail are useful in an applied context, and result in truth-tellers providing a greater quantity of details and providing more consistent accounts than liars (Mac Giolla & Luke, 2021). Frameworks that have achieved relatively consistent research evidence include the Strategic Use of Evidence (SUE) (Granhag, 2010) and the Verifiability Approach (VA) (Nahari et al., 2012). SUE exploits the strategic differences between liars and truth-tellers, inasmuch that liars tend to be avoidant and vague in order to avoid detection, or to use denials. By not initially disclosing the evidence held, investigators can use evidence-related questions to expose a lack of internal consistency and a lack of consistency with the evidence, in deceptive accounts (). The VA is based on the premise that liars try to avoid providing details that can be checked by investigators (as this would result in being caught out), but also want to provide a detailed account as this is considered more believable. The resulting strategy adopted by some liars is therefore to provide a quantity of details that cannot be verified. To magnify this effect, investigators can specifically ask interviewees to provide details that can be checked by the investigator, as truth-tellers are able to do this to a higher degree than liars.

Controlled Cognitive Engagement (CCE) (Ormerod and Dando, 2015) is an interview model that as yet has a modest evidence base, but early findings based on rigorous methodology are promising, particularly in the context of aviation security. The model has three phases, which are repeated in cycles around topics. In the first phase the interviewer asks neutral questions, which are non-challenging for both liars and truth-tellers to answer truthfully. The purpose of this is to develop rapport and to establish a baseline of how an interviewee responds verbally when being truthful. The second phase is information-gathering, in which the interviewee is asked an open question and commits themselves to an account. In the third phase, a question is

asked that tests the veracity of the preceding information; these are focused questions, often about verifiable details, that the interviewee should be able to provide if their previous answers were true. The interviewer looks for changes in the quality of the answers provided; liars produce shorter responses and less information than truth-tellers as the questions continue, they have more inconsistencies in their responses, and more gaps in expected knowledge. Investigative interviewers trained in the PEACE framework will recognise some of the techniques mentioned earlier.

Summary

In summary, the search for Pinocchio's nose is unlikely to ever be wholly successful. Rather than focusing on individual cues, it may be more productive to look for clusters of cues; a meta-analysis achieved relatively high accuracy in classifying communications as truthful or deceptive using combinations of cues (Hartwig & Bond, 2014). For example, several indicators that an interviewee is experiencing increased cognitive load in a particular area of an account. Cognitive approaches and active strategies to maximise the verbal differences between deceptive and truthful accounts have solid evidence bases and are probably most useful in an investigative context as they are non-invasive and can be incorporated into existing practices. Cues that are common to these approaches are a lack of detail and consistency in liars' communications, and these can be useful evidentially. Future technological approaches may centre around the guilty knowledge indicated by the P300 response, although it is unlikely to be generally admissible as evidence in near future.

CLEA WRIGHT

***Relevant chapters:* Expert Evidence; False Allegations and Wrongful Convictions; False Confessions; False Memories; Investigative Interviewing**

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